

Deep Learning

**A Napkin-style companion for campus placements:
from the perceptron to Transformers and LLMs, with derivations,
hand numerics, traps, and code that checks itself**

Philospher

Built October 6, 2026

Preface

[STUB: preface] Versions: 2.14.1+cu130.

Contents

Preface	iii
I Foundations	7
1 Perceptron and the multilayer perceptron	9
1.1 Perceptron and the multilayer perceptron	9
2 Activations, losses and output layers	11
2.1 Activations, losses and output layers	11
3 Backpropagation	13
3.1 Backpropagation	13
4 Optimisers and learning-rate schedules	15
4.1 Optimisers and learning-rate schedules	15
5 Initialisation, normalisation and regularisation	17
5.1 Initialisation, normalisation and regularisation	17
II Architectures	19
6 Convolutional networks	21
6.1 Convolutional networks	21
7 Recurrent networks	23
7.1 Recurrent networks	23
8 Attention and the Transformer	25
8.1 Attention and the Transformer	25
III Modern Deep Learning	27
9 Language models and LLMs	29
9.1 Language models and LLMs	29
10 Generative and representation learning	31
10.1 Generative and representation learning	31
IV Interview Mode	33
11 Practice and PyTorch	35
11.1 Practice and PyTorch	35
12 Implement it	37
12.1 Implement it	37

13	Rapid-fire	39
13.1	Rapid-fire	39
14	Problems	41
14.1	Problems	41
V	Backmatter	43
A	Hints to problems	45
B	Solutions to problems	47

I

Foundations

Part I: Contents

1	Perceptron and the multilayer perceptron	9
1.1	Perceptron and the multilayer perceptron	9
2	Activations, losses and output layers	11
2.1	Activations, losses and output layers	11
3	Backpropagation	13
3.1	Backpropagation	13
4	Optimisers and learning-rate schedules	15
4.1	Optimisers and learning-rate schedules	15
5	Initialisation, normalisation and regularisation	17
5.1	Initialisation, normalisation and regularisation	17

1 Perceptron and the multilayer perceptron

§1.1 Perceptron and the multilayer perceptron

[STUB: Perceptron and the multilayer perceptron]

2 Activations, losses and output layers

§2.1 Activations, losses and output layers

[STUB: Activations, losses and output layers]

3 Backpropagation

§3.1 Backpropagation

[STUB: Backpropagation]

4 Optimisers and learning-rate schedules

§4.1 Optimisers and learning-rate schedules

[STUB: Optimisers and learning-rate schedules]

5 Initialisation, normalisation and regularisation

§5.1 Initialisation, normalisation and regularisation

[STUB: Initialisation, normalisation and regularisation]

II

Architectures

Part II: Contents

6	Convolutional networks	21
6.1	Convolutional networks	21
7	Recurrent networks	23
7.1	Recurrent networks	23
8	Attention and the Transformer	25
8.1	Attention and the Transformer	25

6 Convolutional networks

§6.1 Convolutional networks

[STUB: Convolutional networks]

7 Recurrent networks

§7.1 Recurrent networks

[STUB: Recurrent networks]

8 Attention and the Transformer

§8.1 Attention and the Transformer

[STUB: Attention and the Transformer]

III

Modern Deep Learning

Part III: Contents

9	Language models and LLMs	29
9.1	Language models and LLMs	29
10	Generative and representation learning	31
10.1	Generative and representation learning	31

9 Language models and LLMs

§9.1 Language models and LLMs

[STUB: Language models and LLMs]

10 Generative and representation learning

§10.1 Generative and representation learning

[STUB: Generative and representation learning]

IV

Interview Mode

Part IV: Contents

11	Practice and PyTorch	35
11.1	Practice and PyTorch	35
12	Implement it	37
12.1	Implement it	37
13	Rapid-fire	39
13.1	Rapid-fire	39
14	Problems	41
14.1	Problems	41

11 Practice and PyTorch

§11.1 Practice and PyTorch

[STUB: Practice and PyTorch]

12 Implement it

§12.1 Implement it

[STUB: Implement it]

13 Rapid-fire

§13.1 Rapid-fire

[STUB: Rapid-fire]

14 Problems

§14.1 Problems

[STUB: Problems]



Backmatter

A Hints to problems

B Solutions to problems